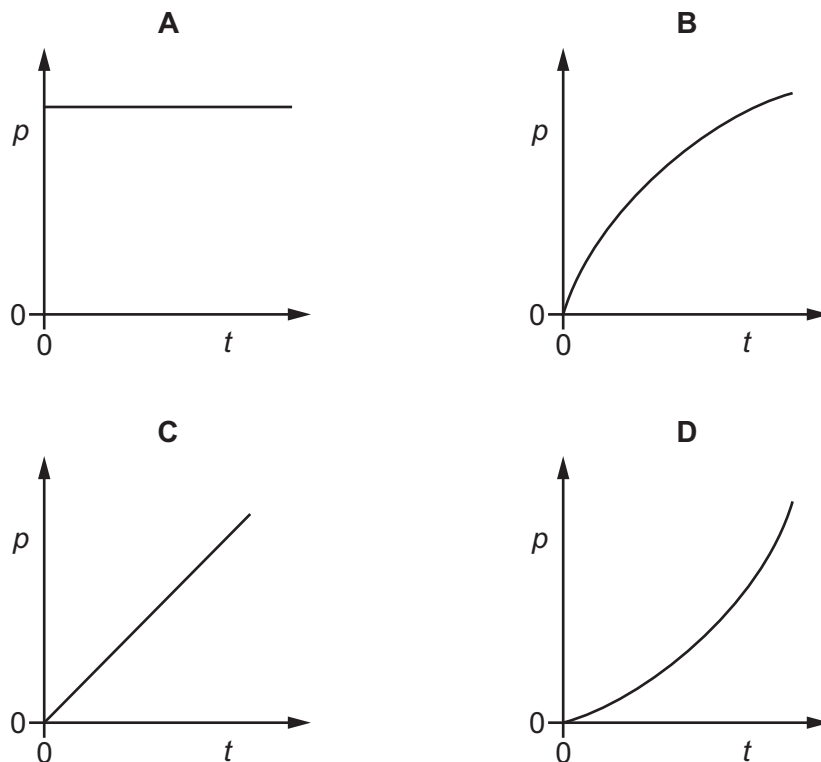


- 4 An object falls from rest towards the ground.

Air resistance is negligible.

Which graph shows the variation of the momentum p of the object with time t until it hits the ground?



- 5 A projectile is fired at an angle of 45° upwards from horizontal ground. Air resistance is negligible.

Which row describes the horizontal motion and the vertical motion of the projectile after it is fired until immediately before it reaches the ground again?

	horizontal motion	vertical motion
A	constant velocity	constant acceleration
B	constant velocity	varying acceleration
C	varying velocity	constant acceleration
D	varying velocity	varying acceleration

- 6 An aircraft on a runway accelerates uniformly from rest to its take-off speed of 58 m s^{-1} .

The acceleration of the aircraft is 4.2 m s^{-2} , and the aircraft uses 74% of the length of the runway to reach its take-off speed.

What is the length of the runway?

- A** 300 m **B** 540 m **C** 590 m **D** 800 m